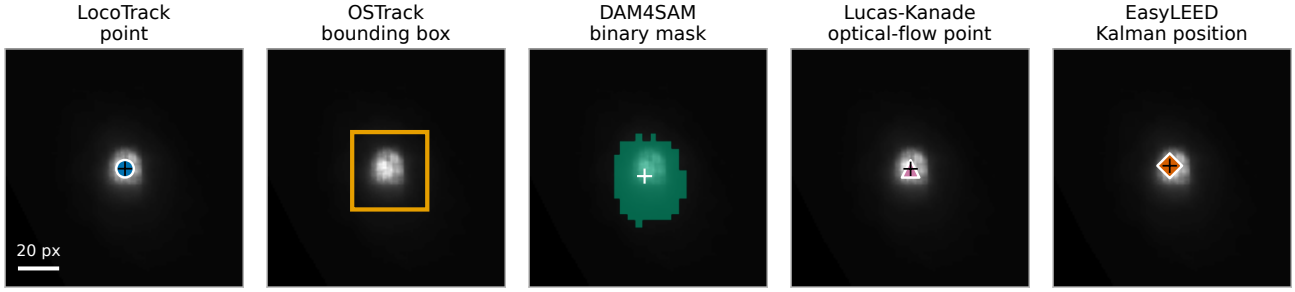
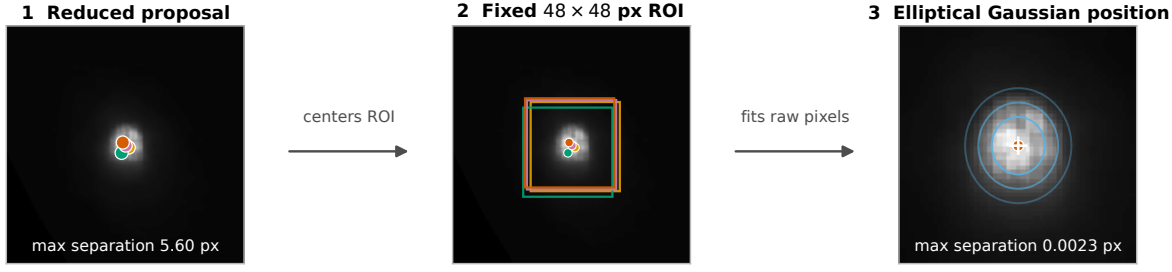
a Heterogeneous native outputs for the same Cu(111) spot

56 eV, 92 °C



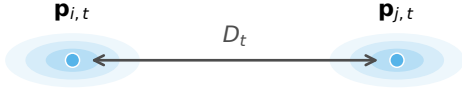
Empirical point, bbox, mask, optical-flow point and Kalman position; same image and spot identity.

b Common proposal-to-measurement conversion



The tracker locates the measurement region; the common Gaussian fit defines the quantitative spot position.

c Pairwise physical endpoint



$$D_t = \|\mathbf{p}_{i,t} - \mathbf{p}_{j,t}\|$$

$$\Delta D/D = (D_t - D_{\mathrm{ref}})/D_{\mathrm{ref}}$$

Common image translation cancels; non-rigid distortion does not.

d Four complementary evidence roles

1. **Positive control** — Cu(111) thermal sequence
2. **Equivariance** — Known translations
3. **Motion + I(V) gaps** — Ag(111) energy scans
4. **Memory mechanism** — SAM2 family

No pooled score: separate questions and denominators.